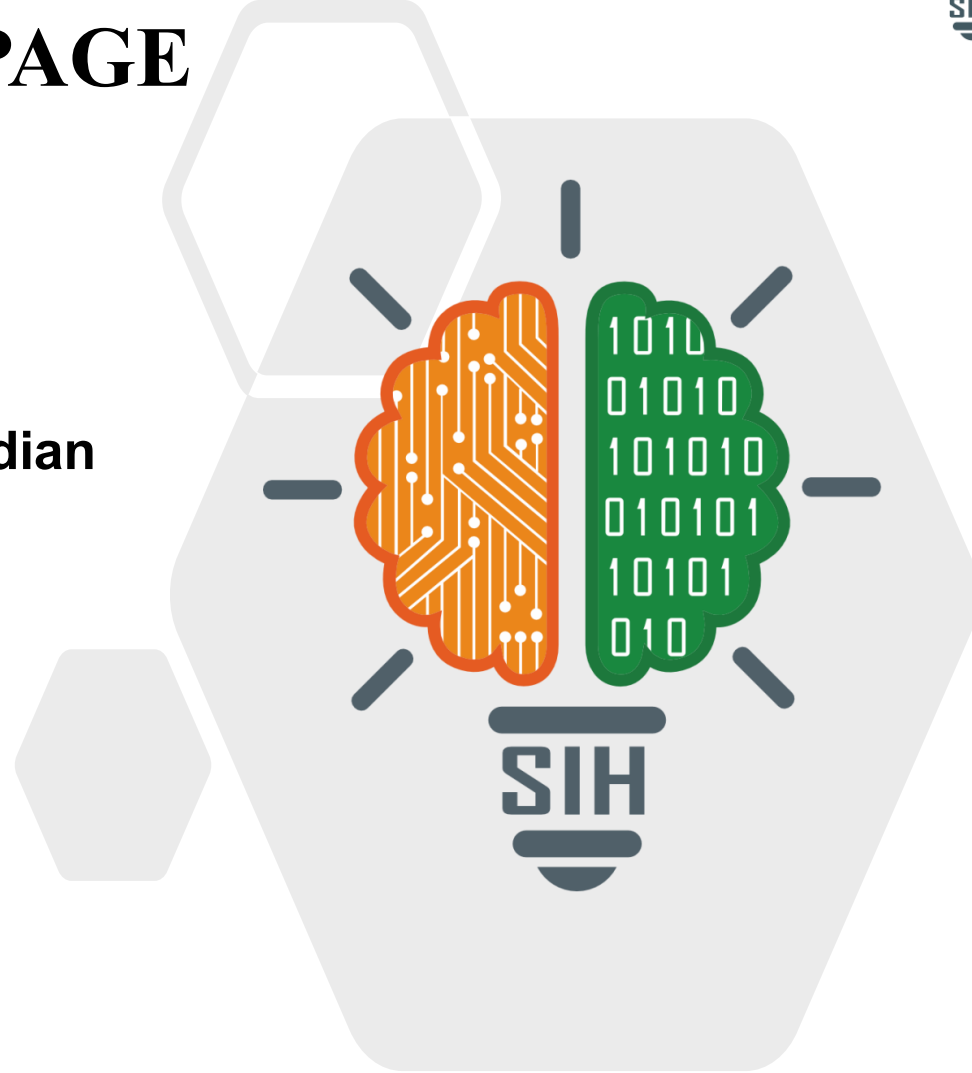


TITLE PAGE

- Problem Statement ID – SIH26037
- Problem Statement Title – Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- Theme – Smart Vehicles
- PS Category – Software
- Team ID – 161638
- Team Name – Ctrl Alt Elite
- Solution – PathSense



PATHSENSE

Adaptive Path Planning & Collision Avoidance for Unstructured Indian Roads



PROPOSED SOLUTION

PathSense is an adaptive autonomous-driving intelligence system for Indian roads where lane markings may be absent and traffic behaviour is irregular. **Core idea:** perceive every road user, understand its motion, reason about the vehicle's real drivable corridor and keep re-planning a collision-free trajectory.



PROBLEMS IT ADDRESSES

Unmarked Roads

Lane markings are faded, partial or absent, so lane-following assumptions break.

Mixed Traffic

Cars, buses, trucks, two-wheelers, auto-rickshaws, cyclists and pedestrians share one space.

Informal Merges

Vehicles cut in from either side without signalling or lane discipline.

Sudden Direction Changes

Two-wheelers and autos weave, U-turn or stop abruptly.

Wrong-Way Movement

Riders and vehicles travel against the flow on the vehicle's own side.

Pedestrian / Cyclist Interaction

People walk along and across the carriageway without crossings.

Animal Crossings

Cattle and dogs stand on, or wander into, the road.

Temporary Obstacles

Work zones, parked vehicles, handcarts and debris block the usual path.

DIFFERENTIATION & KEY VALUE PROPOSITION

01 Lane-Independent Path Planning

Plans around the actual drivable corridor instead of depending only on painted lane markings.

02 Indian Mixed-Traffic Awareness

Designed for cars, buses, trucks, motorcycles, auto-rickshaws, pedestrians, bicycles and animals.

03 Behaviour-Aware Collision Avoidance

Considers object movement and potential path conflict rather than distance alone.

04 Adaptive Replanning

Continuously evaluates candidate trajectories as the road scene changes.

05 Unstructured-Road Operation

Targets roads where conventional lane-based assumptions are unreliable.

06 Safety-First Decision Layer

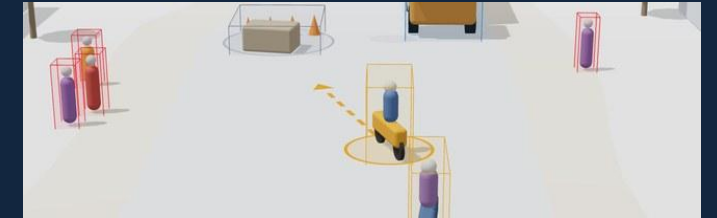
Escalates from normal travel to caution, steering, braking and no-safe-path according to path threat.

How It Works

From the camera view to a safe, explainable path

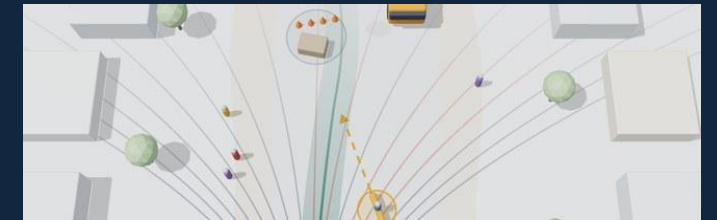
1 Perceive & Track

Every road user, every frame



2 Predict & Plan

Candidate paths, conflicts rejected



3 Decide

GO · SLOW · STEER · BRAKE



SEE · PREDICT · PLAN — SAFE PATHS ON EVERY ROAD

TECHNICAL APPROACH

PathSense | Adaptive Perception-to-Planning Pipeline



INPUTS / ROAD SCENE



Front camera video

Monocular dashcam or phone



Mixed traffic

Cars, 2W, autos, people, animals



Road context

Unmarked roads, shoulders, junctions



Ego motion

From video; CAN / IMU in pilot

1 · PERCEIVE

1



Camera input

front view, every frame

2



Object detection

vehicles, people, animals

3



Multi-object tracking

IDs & history

2 · UNDERSTAND

4



Depth & scene

distance, free space

5



Motion / threat

who is moving where

6



Ego-path & corridor

real drivable corridor

3 · PLAN & ACT

7



Candidate trajectories

kinematic path set

8



Risk-aware selection

reject conflicts

9



Steering / decision

path + GO ... BRAKE


Safety-First Decision Layer path-threat reasoning | persistence & hysteresis | reasons

OUTPUTS / DECISIONS



Safe trajectory

Selected collision-free path



Driving decision

GO · SLOW · STEER · BRAKE · NO SAFE PATH



Threat alerts

Which road user, why, how soon



Decision log

Explainable, replayable reasons

Designed for:

Urban dense traffic

crowds, vendors, turning buses

Village roads

narrow, unmarked, cattle, tractors

Highways & transitions

merges, trucks, higher speeds

Unmarked city roads

work zones, wrong-way riders

TECH STACK

Perception



YOLO



OpenCV

Tracking



ByteTrack

Depth / Vision



Depth Anything



CLIP

Planning



Python



NumPy

Web / Demo



Flask



HTML



CSS



JS

Runtime



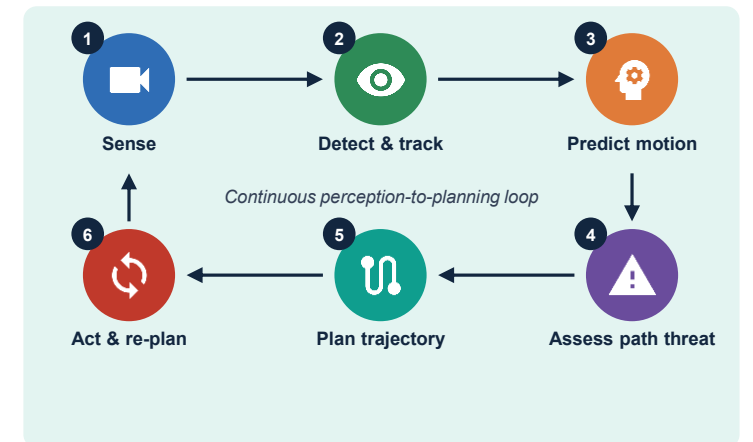
PyTorch



CUDA

GPU-accelerated inference with open pretrained models

METHODOLOGY

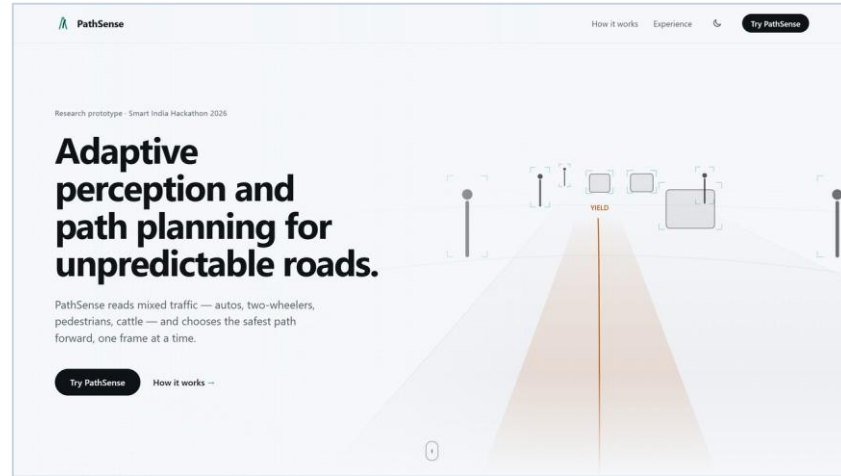


PROTOTYPE

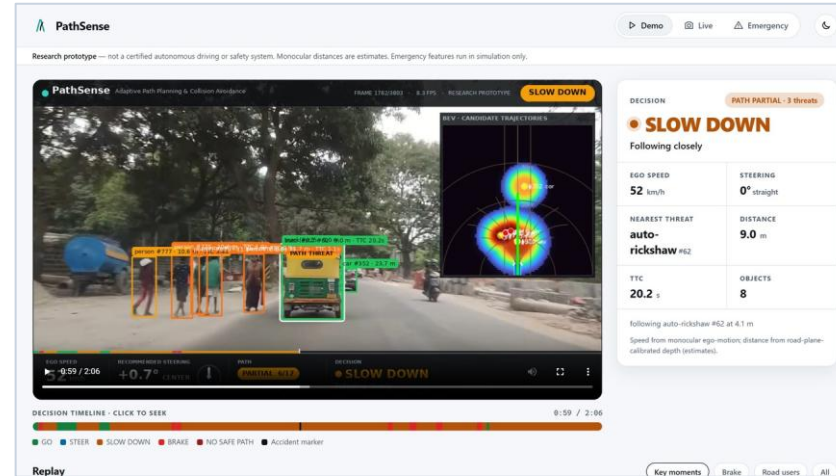
Screens from the working PathSense web application (research prototype)



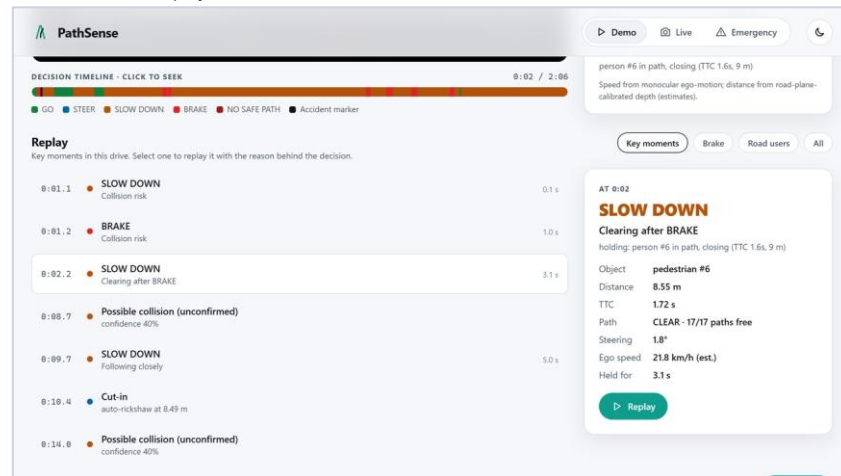
01 Landing page Entry point; "Try PathSense" opens the app (Demo · Live · Emergency)



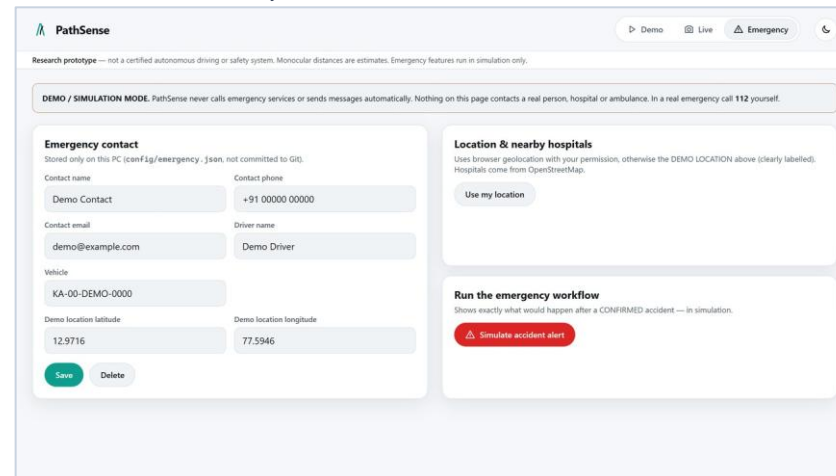
02 Demo playback Processed drive: detections, bird's-eye candidate trajectories, decision and reason



03 Replay of key moments Each moment lists object, distance, TTC and path status; click to replay



04 Emergency (simulation mode) Confirmation-first workflow; never contacts real services automatically



Current Development

Working prototype: camera-based perception, tracking, motion understanding and adaptive path planning

Code repository:

github.com/MalyalaKarthik66/pathsense-prototype

Live demonstration shown separately during evaluation.

What the prototype demonstrates

- Detection & tracking of mixed Indian road users
- Path-threat reasoning: oncoming vs. entering the path
- Adaptive candidate-path planning around obstacles
- Explainable decisions with replay

FEASIBILITY AND VIABILITY



✓ FEASIBILITY

- **Technology:** Established computer-vision and motion-planning building blocks; camera-based perception to start.
- **Architecture:** Modular perception → prediction → planning; radar / LiDAR and vehicle-state inputs added progressively; scenario-based simulation supports development.

🚀 VIABILITY

- **Relevance:** Mixed, irregular traffic on urban roads, village roads, highways and dense traffic.
- **Deployment:** Modular architecture supports incremental deployment through controlled pilots.
- **Future fit:** Can integrate with future intelligent-vehicle platforms.

RISKS → MITIGATION

RISK	MITIGATION
⚠️ Unstructured road geometry	→ Drivable-corridor estimation and path-aware planning.
⚠️ Unpredictable road-user behaviour	→ Temporal tracking and short-term motion reasoning.
⚠️ Perception uncertainty	→ Confidence-aware decisions and future sensor fusion.
⚠️ Real-world validation complexity	→ Scenario library + simulation + controlled field testing.

PROTOTYPE → PILOT → PRODUCTION

1 PROTOTYPE

Camera-based proof of concept for perception, tracking and adaptive path planning.

2 PILOT

Multi-scenario validation with richer sensing, simulation and controlled road testing.

3 PRODUCTION

Vehicle-grade multi-sensor integration, closed-loop validation and automotive safety engineering.

POTENTIAL COLLABORATORS / PARTNERS

Automotive OEMs & Tier-1s ADAS / AV integration	Research institutes IITs, IISc, AV labs
Simulation partners e.g. MathWorks toolchain	Road-safety bodies MoRTH, state transport
Fleet operators Buses, taxis, logistics	Mapping & data OpenStreetMap, IDD

PROOF OF CONCEPT, IMPACT & BENEFITS



PROOF OF CONCEPT

1

PROBLEM

Unstructured Indian roads create difficult path-planning conditions

2

OBJECTIVE

Safe adaptive trajectories without relying solely on lane markings

3

PROTOTYPE

Camera-based perception, tracking, motion understanding and adaptive planning

4

VALIDATION

Scenario-based evaluation across mixed-traffic road situations

5

SUCCESS CRITERIA

Safe path selection, appropriate risk response, robust behaviour across roads

6

NEXT STEPS

Sensor fusion, closed-loop simulation, broader field trials, vehicle integration

Resources: computer vision & motion planning, road-scene video, simulation tools, compute hardware

POTENTIAL IMPACTS

Target Audience



Drivers / Passengers: Improved handling of unpredictable road interactions



Urban Mobility: More adaptive navigation in dense mixed traffic



Rural Mobility: Better operation on roads with limited lane structure



Road Safety: Earlier recognition of path conflicts and adaptive response



Future Autonomous Vehicles: A foundation for Indian-road-aware navigation

BENEFITS



SAFETY

- Path-aware collision avoidance
- Earlier recognition of path conflicts



ADAPTABILITY

- Handles changing road and traffic conditions
- Works where lane markings are unreliable



SCALABILITY

- Can evolve from prototype → pilot → production
- Modular: new sensors and platforms plug in



INDIAN-ROAD RELEVANCE

- Designed around mixed and irregular road behaviour
- Safer shared roads for pedestrians and two-wheelers

BUSINESS MODEL CANVAS



Key Partnerships

- **OEMs:** Automotive OEMs & Tier-1 ADAS suppliers
- **ACADEMIA:** IITs, IISc & autonomy research labs
- **SIMULATION:** MathWorks toolchain (RoadRunner, Automated Driving Toolbox)
- **DATA:** Indian driving datasets (IDD), OpenStreetMap
- **GOVT:** MoRTH, state transport & road-safety bodies
- **FLEETS:** Logistics, taxi & public-transport operators

Key Activities

- **PERCEPTION R&D:** Detection, tracking & depth
- **PLANNING:** Adaptive trajectory & risk logic
- **VALIDATION:** Scenario library & simulation
- **INTEGRATION:** Vehicle / ADAS interfaces

Key Resources

- **SOFTWARE STACK:** Perception-to-planning pipeline
- **DATA:** Indian-road scenario library
- **TEAM:** CV, robotics & software engineers
- **TOOLS:** Simulation & compute infrastructure

Value Propositions

- **OEMs:** An Indian-road-aware planning layer
- **FLEETS:** Fewer conflicts in mixed traffic
- **ROAD USERS:** Earlier recognition of path conflicts
- **RESEARCHERS:** Scenario-based validation workflow
- **EXPLAINABLE:** Every decision comes with its reason
- **MODULAR:** Camera-first, sensor-ready architecture

Customer Relationships

- **CO-DEVELOPMENT:** Pilot programmes with OEMs
- **SUPPORT:** Integration & tuning
- **UPDATES:** Continuous model & scenario updates
- **TRANSPARENCY:** Decision logs for audit

Channels

- **DIRECT B2B:** OEM & Tier-1 partnerships
- **PILOTS:** Fleet & campus deployments
- **RESEARCH:** Academic collaborations
- **SDK / API:** Licensable software modules

Customer Segments

- **PRIMARY:** Automotive OEMs & ADAS suppliers
- **SECONDARY:** Fleet & logistics operators
- **PUBLIC:** Transport & road-safety agencies
- **RESEARCH:** AV labs & testing bodies
- **SCALE-UP:** City → state → national deployments

Cost Structure

- **R&D** – perception & planning engineering
- **Data** – collection, annotation & scenario building
- **Compute** – training & simulation infrastructure
- **Validation** – field testing & safety engineering
- **Operations** – integration, support & updates

Revenue Streams

- **Software licensing** – per-vehicle / per-platform licences for OEMs
- **Pilot contracts** – fleet & campus deployments
- **Scenario-library access** – validation datasets & tools
- **Integration services** – customisation & calibration
- **Support & updates** – annual maintenance subscriptions

RESEARCH AND REFERENCES



SUPPORTING RESEARCH PAPERS

AUTONOMOUS DRIVING & PATH PLANNING

"A Survey of Motion Planning and Control Techniques for Self-Driving Urban Vehicles"

Paden et al. — IEEE T-IV, 2016

arxiv.org/abs/1604.07446

DEPTH ESTIMATION

"Depth Anything V2"

Yang et al. — NeurIPS, 2024

arxiv.org/abs/2406.09414

OBJECT DETECTION & TRACKING

"ByteTrack: Multi-Object Tracking by Associating Every Detection Box"

Zhang et al. — ECCV, 2022

arxiv.org/abs/2110.06864

MOTION PREDICTION

"Social LSTM: Human Trajectory Prediction in Crowded Spaces"

Alahi et al. — CVPR, 2016

openaccess.thecvf.com

 INDIAN ROAD DATA

IDD – Indian Driving Dataset

idd.insaan.iit.ac.in

MoRTH – Road Accidents in India

morth.nic.in

OpenStreetMap

openstreetmap.org

Wikimedia Commons

commons.wikimedia.org

 SIH & CONTEXT

SIH 2026 – PS SIH26037

sih.gov.in

MathWorks – RoadRunner & ADT

mathworks.com

Problem focus: unmarked roads, mixed traffic, informal merges, obstacles.

 TECHNICAL DOCS

Ultralytics YOLO

docs.ultralytics.com

PyTorch

pytorch.org/docs

OpenCV

docs.opencv.org

Flask

flask.palletsprojects.com

NATIONAL ALIGNMENT



Road Safety MoRTH road-safety goals

Earlier recognition of path conflicts on Indian roads



IndiaAI Mission AI for Indian conditions

Perception and planning built for Indian traffic



Atmanirbhar Bharat Self-reliant technology

Home-grown autonomous-driving intelligence



Make in India Indigenous automotive tech

ADAS software developed for Indian vehicles



Smart Cities Mission Intelligent urban mobility

Adaptive navigation in dense city traffic



Digital India Software-first innovation

Open, modular and explainable mobility software

Sources: IEEE T-IV · ECCV · NeurIPS · CVPR · IDD (IIIT-H) · MoRTH · OpenStreetMap · SIH 2026 · Ultralytics · PyTorch · OpenCV